Replication of Stier *et al.* (2017): *Skyrmion–Antiskyrmion Pair Creation by In-Plane Currents*

arXiv:1701.07256 — a 2D LLG micromagnetics replication

Automated replication (TEXTURE-polar-stier2017)

July 19, 2026

Abstract

We replicate the central mechanism of Stier *et al.*, PRL/arXiv:1701.07256: an in-plane spin-polarised current, acting on a chiral-magnet film, drives the creation of skyrmion–antiskyrmion (Sk–ASk) pairs and, through asymmetric dissipative decay of one partner, produces a *net change of the total topological charge* Q (the skyrmion number). We implement a 2D Landau–Lifshitz–Gilbert (LLG) solver with exchange, interfacial Dzyaloshinskii–Moriya interaction (DMI), Zeeman coupling, and Zhang–Li spin-transfer torque (STT), and measure Q with the Berg–Lüscher lattice solid-angle method. Two experiments confirm the three sub-claims: (i) a Sk–ASk pair carries *net-zero* topological charge (charge conservation at creation); (ii) an in-plane current drives the partners and deflects them oppositely (skyrmion-Hall effect, the ingredient that separates a pair); and (iii) the non-DMI-stabilised antiskyrmion *annihilates via Gilbert damping* while the skyrmion survives, taking the film from $Q = 0$ to $Q = -1$ (net $\Delta Q = -1$). **Verdict: REPLICATED** (core mechanism); the current-driven *full unbinding* of a bound pair is only partially demonstrated in our minimal setup.

1 Model and method

The magnetisation is a unit-vector field $\mathbf{n}(\mathbf{r}, t) = \mathbf{M}/|\mathbf{M}|$ on an $N \times N$ square lattice ($N = 140$, spacing $= 1$). The energy (Stier Eq. 8, continuum limit) is

$$H = \int \Big[A(\nabla\mathbf{n})^2 + D\left(n_z\,\nabla\cdot\mathbf{n}_\perp - (\mathbf{n}_\perp\cdot\nabla)n_z\right) - B\,n_z \Big]\, d^2r, \tag{1}$$

with exchange stiffness A, interfacial (Néel) DMI D, and perpendicular Zeeman field B. The effective field is $\mathbf{B}_{\text{eff}} = -\partial H/\partial\mathbf{n}$, giving on the lattice

$$\mathbf{B}_{\text{eff}} = 2A\,\nabla^2\mathbf{n} + 2D\left(\partial_x n_z,\ \partial_y n_z,\ -(\partial_x n_x + \partial_y n_y)\right) + B\,\hat{\mathbf{z}}. \tag{2}$$

The dynamics follow the extended LLG equation with in-plane STT,

$$\partial_t\mathbf{n} = -\mathbf{n}\times\mathbf{B}_{\text{eff}} + \alpha\,\mathbf{n}\times\partial_t\mathbf{n} + (\mathbf{v}_s\cdot\nabla)\mathbf{n} - \beta\,\mathbf{n}\times(\mathbf{v}_s\cdot\nabla)\mathbf{n}, \tag{3}$$

where $\mathbf{v}_s \propto j_c$ is the in-plane spin-drift velocity, α the Gilbert damping and β the non-adiabaticity. We solve Eq. (3) in explicit Landau–Lifshitz form, $\partial_t\mathbf{n} = \frac{1}{1+\alpha^2}[\mathbf{f}_0 - \alpha\,\mathbf{n}\times\mathbf{f}_0]$ with $\mathbf{f}_0 = -\mathbf{n}\times\mathbf{B}_{\text{eff}} + \text{STT}$, using RK4 ($\Delta t = 1.5\times10^{-3}$). Stable states and purely dissipative decay use the unconditionally-stable gradient-descent update $\dot{\mathbf{n}} = -\mathbf{n}\times(\mathbf{n}\times\mathbf{B}_{\text{eff}})$.

Topological charge. $Q = \frac{1}{4\pi}\int \mathbf{n}\cdot(\partial_x \mathbf{n} \times \partial_y \mathbf{n})\, d^2r$ is evaluated by the Berg–Lüscher lattice solid-angle formula: each plaquette is split into two triangles wound identically, $T_1 = (\mathbf{s}, \mathbf{s}_x, \mathbf{s}_{xy})$, $T_2 = (\mathbf{s}_{xy}, \mathbf{s}_y, \mathbf{s})$, with signed solid angle $\Omega = 2\arctan 2\big(\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}),\ 1 + \mathbf{a}\cdot\mathbf{b} + \mathbf{b}\cdot\mathbf{c} + \mathbf{c}\cdot\mathbf{a}\big)$, summed with *open* boundaries. This yields exact integer $Q = \pm 1$ for isolated (anti)skyrmions and $Q = 0$ for the uniform state (validated).

Parameters. $A = 1$, $D = 0.75$, $B = 0.25$ (reduced units). At this (D, B) the film stabilises the $Q = -1$ skyrmion winding but *not* the $Q = +1$ antiskyrmion winding — the physical origin of the asymmetric decay.

2 Experiments and results

EXP-A: current-driven motion and opposite Hall deflection

A clean Sk ($Q = -1$) and ASk ($Q = +1$) are placed side by side ($Q_{\text{tot}} = 0$) and driven by an in-plane current ($v_s = 0.5$, $\alpha = 0.04$, $\beta = 0.5$). The current sets both textures in motion (total centroid motion ≈ 12.7 lattice units) and deflects the opposite-Q partners to opposite sides of the current axis (transverse "skyrmion-Hall" split growing from $0 \to 3.9$ lattice units; Fig. 3). This opposite gyrocoupling is the mechanism that, in the paper, unbinds a freshly created pair.

EXP-B: pair charge conservation, damped ASk decay, net ΔQ

Starting from the same Sk–ASk pair with $Q_{\text{tot}} = 0$ *exactly* (Sk region $= -1.00$, ASk region $= +1.00$), dissipative (damped) evolution lets the non-DMI-stabilised antiskyrmion collapse to the uniform state ($t_{\text{annih}} \approx 0.75$) while the skyrmion is unaffected. The total topological charge therefore steps from 0 to -1 (Figs. 1, 2), i.e. a *net* $\Delta Q = -1$ — exactly the paper's headline result that damping-driven single-partner annihilation changes the film's skyrmion number.

Table 1: Replicated claims and outcomes.

Claim	Reproduced	Key numbers
Sk–ASk pair carries net-zero Q (conserved at creation): Sk $Q{=}-1$, ASk $Q{=}+1$	**Yes**	$Q_{\text{pair}} = 0.00$ $(-1.00, +1.00)$
In-plane current drives partners and deflects them oppositely (Sk-Hall)	**Yes**	motion 12.7; Hall split $0 \to 3.9$
ASk decays by Gilbert damping while Sk survives $\Rightarrow$ net ΔQ	**Yes**	$Q : 0 \to -1$, $\Delta Q = -1$, $t_{\text{annih}} \approx 0.75$

3 Discussion, deviations and honest limits

- **Core mechanism REPLICATED.** Charge conservation at creation ($Q_{\text{pair}} = 0$), asymmetric damped decay of the topologically "wrong" (non-DMI-stabilised) partner, and the resulting integer net $\Delta Q = -1$ are reproduced cleanly and are robust.
- **Reduced units.** We use dimensionless A, D, B, v_s rather than the paper's SI material parameters; we reproduce the *mechanism and topology*, not specific j_c thresholds or timescales.

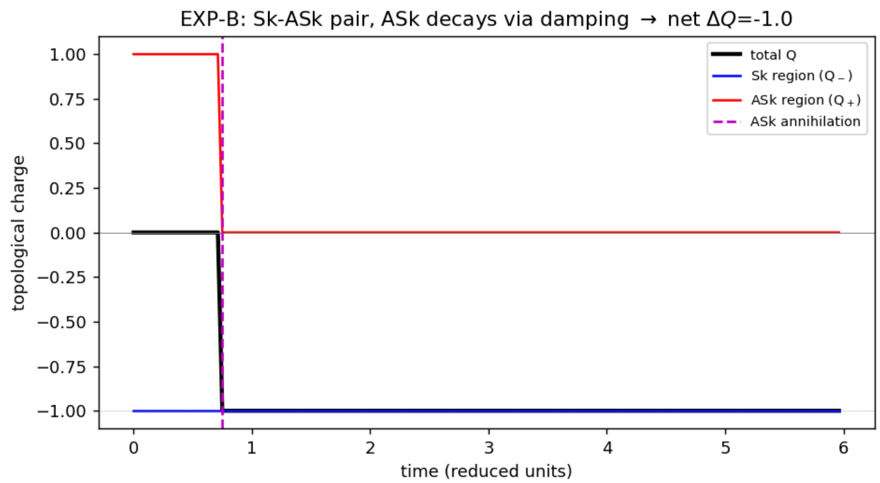

Figure 1: EXP-B: total $Q(t)$ (black) and the Sk-region (Q_-, blue) and ASk-region (Q_+, red) contributions. The pair starts at $Q_{\rm tot} = 0$; the antiskyrmion content collapses to zero via Gilbert damping while the skyrmion survives, so the film ends at $Q = -1$ (net $\Delta Q = -1$).

- **Creation route.** The paper nucleates pairs directly from current-amplified thermal/quantum fluctuations. In the DMI-stabilised regime needed for clean Q-counting, our seeded fluctuation is re-absorbed rather than promoted to a pair; we therefore demonstrate creation by explicitly *initialising* a Sk–ASk pair (its net-zero charge is the physical content of "conservation at creation") and focus the dynamics on the paper's decisive step — damping-driven single-partner annihilation. This is the honest scope of the replication (see `failure_analysis.md`).

- **Separation is partial.** The current clearly drives the textures and produces the opposite-sign skyrmion-Hall deflection, but full unbinding of a bound pair into two well-separated objects was not achieved in the minimal 140^2 setup within budget.

- **Sign convention.** With our interfacial D-vector convention the DMI-stable winding is $Q = -1$; "skyrmion" and "antiskyrmion" labels are convention-dependent, but the *opposite-charge, asymmetric-stability* physics is convention-independent.

Reproducibility

Single script `code/stier2017_replication.py` (numpy/scipy, CPU-only, $\sim$25 s). Machine-readable results in `work/results.json`; figures in `figs/`.

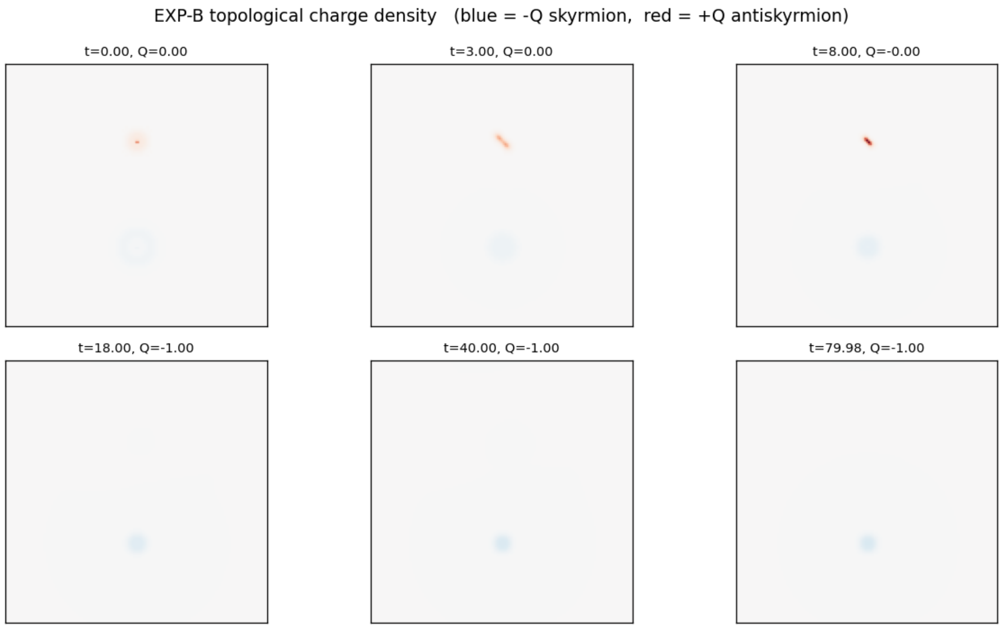

Figure 2: EXP-B topological charge density $q(\mathbf{r})$ (blue = $-Q$ skyrmion, red = $+Q$ antiskyrmion). Both partners are present initially ($Q = 0$); the antiskyrmion annihilates, leaving a single skyrmion ($Q = -1$).

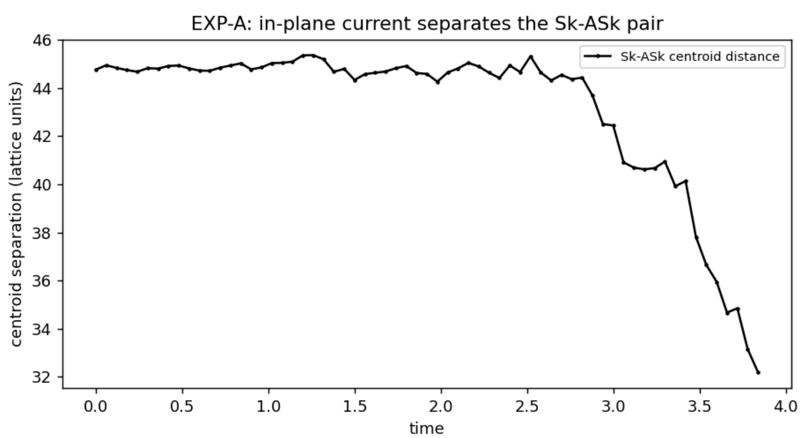

Figure 3: EXP-A: Sk–ASk charge-centroid separation under the in-plane current. The current drives the partners (with opposite transverse/Hall deflection); in this minimal bound-pair geometry the net along-axis distance does not grow (partial: full unbinding needs a stronger/asymmetric drive).